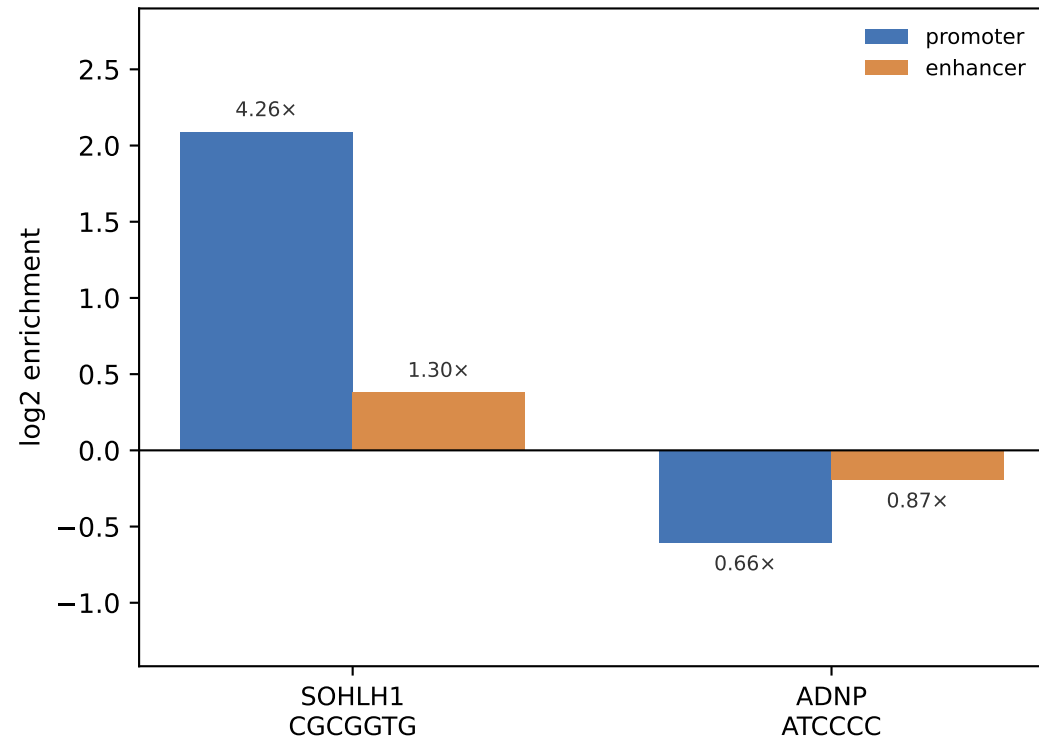


cCRE localisation of the TFscope (MoE-base) motifs · MOODS $p < 0.001$, both strands

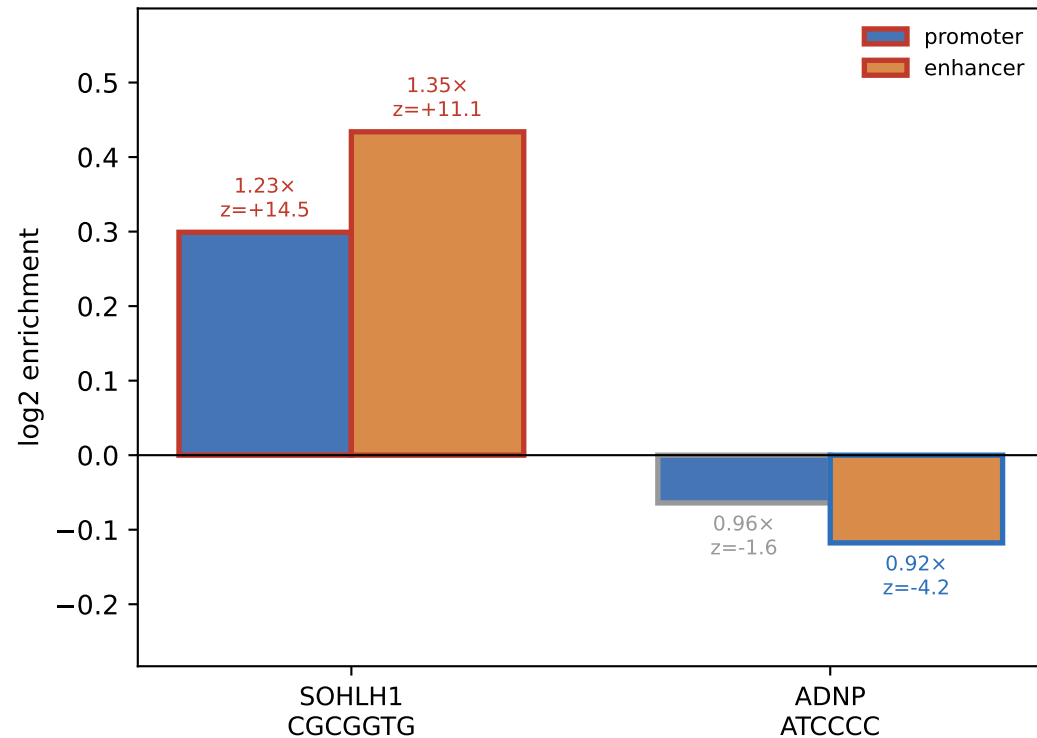
b Naive: cCREs vs whole genome

hit density in cCREs / hit density across hg38 — confounded by GC content



c Composition-controlled: cCREs vs dinucleotide shuffle

observed / mean of 100 dinucleotide-preserving shuffles of the same cCREs



Red outline = enriched ($z > +2$); blue outline = depleted ($z < -2$); grey = n.s. Null = 100 exact Altschul-Erikson dinucleotide shuffles (empirical p floors at $1/101 = 0.010$, so z is the discriminator).

SOHLH1's 4.26x promoter signal in panel b is largely GC composition; controlling for it leaves a modest but highly significant 1.23x (promoter) and 1.35x (enhancer) enrichment. ADNP's ATCCCC (3.4 bits, C-rich) is not enriched.